EXPERIMENT-2

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Academic Year: 2025-2026

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sandeepsabharwal submit experiment 12ba7aa · 3 weeks ago

429 lines (429 loc) · 8.73 KB

Preview Code Blame Raw

1. Declare these variables (x, y, and z) as integers. Assign a value of 9 to x. Assign a value of 7 to y, perform addition, multiplication, division and subtraction on these two variable and print out the results

In [ ]:

```
x = 9
y = 7
print("Addition:",x+y)
print("Multiplication", x*y)
print("Division", x/y)
print("Subtration", x-y)
```

```
Addition: 16
Multiplication 63
Division 1.2857142857142858
Subtration 2
```

2. Write a program where the radius is taken as input to compute the area of a circle

In [ ]:

```
r = int(input("Enter the radius of a circle"))
area = 3.14 * r * r
print("Area of the circle:",area)
```

```
Area of the circle: 28.259999999999998
```

3. Write a python program to solve (x+y)*(x+y)

In [ ]:

```
x = int(input("Enter the value of x"))
y = int(input("Enter the value of y"))
z = (x+y)*(x+y)
print(z)
```

```
49
```

6. Write a program to compute the length of the hypotenuse (c) of a right triangle using Pythagoras theorem

In [ ]:

```
import math
h = int(input("Enter the height of the triangle"))
b = int(input("Enter the base of the triangle"))
sum = (h * h) + (b * b)
c = math.sqrt(sum)
print(c)
```

```
5.0
```

7. Write a program to find simple interest

In [ ]:
```
p = int(input("Enter the principle"))
t = int(input("Enter the time of the loan"))
r = int(input("Enter the rate of interest"))
si = (p * r * t)/100
print(si)
```

```
3000.0
```

8. Write a program to find the area of triangle when lengths of sides are given

In [ ]:
```
import math
a = float(input("Enter side a: "))
b = float(input("Enter side b: "))
c = float(input("Enter side c: "))
s = (a + b + c) / 2
area = math.sqrt(s * (s - a) * (s - b) * (s - c))
print(area)
```

```
0.0
```

9. Write a program to convert given seconds into hours, minutes and remaining seconds.

In [ ]:
```
time = int(input("Enter the seconds"))
hour = time // 3600
rem_time = time % 3600
min = rem_time // 60
sec = rem_time % 60
print(hour, "hour", min, "minutes", sec, "seconds")
```

```
1 hours 34 minutes 38 seconds
```

10. Write a program to swap two numbers without takinf additional variable.

In [5]:
```
a = int(input("Enter the first number"))
b = int(input("Enter the second number"))
a = a + b
b = a - b
a = a -b
print(a,b)
```

```
9 5
```

11. Write a program to find the sum of first n natural numbers.

In [8]:
```
n = int(input("Enter the value to find the sum of nmbers till that number"))
sum = n * (n+1) // 2
print(sum)
```

```
15
```

12. Write a program to print truth table for bitwise operators (&, |, and ^ operators)

In [9]:

```
print("A B  A&B  A|B  A^B")
print("-------------------")
for a in range(2):
    for b in range(2):
        and_op = a & b
        or_op  = a | b
        xor_op = a ^ b

        print(a, b, "  ", and_op, "  ", or_op, "  ", xor_op)
```

```
A B  A&B  A|B  A^B
-------------------
0 0    0    0    0
0 1    0    1    1
1 0    0    1    1
1 1    1    1    0
```

13. Write a program to find the left shift and right shift values of a given number

In [ ]:

```
num = int(input("Enter a number"))
shift = int(input("Enter the number of shifts"))
left_shift = num << shift
right_shift = num >> shift
print(left_shift)
print(right_shift)
```

```
177776
2777
```

14. Using membership operator find whether a given number is in sequence (10,20,56,78,89)

In [13]:

```
num = [10,20,56,78,89]
print(10 in num)
```